

The Role of Digital Twins in Energy Optimization and Smart Building Management: A Systematic Review using PRISMA

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Abstract

Digital Twins (DT) have emerged as enablers of energy optimization in Smart Buildings, but the research landscape remains fragmented. Following PRISMA 2020 across four databases (2015–2025), an initial corpus of 704 records was reduced through a multi-stage LLM-assisted screening pipeline validated by all five team members, yielding 45 included studies. The corpus is dominated by Energy consumption & supervision (71 %); 71 % implement a true Twin (Kritzinger taxonomy); reported energy savings range 15–33 % (median $\approx$ 22 %); an objective technical mapping (RQ4) identifies EnergyPlus, Python and Model Predictive Control as the dominant stack, while OPC-UA and Monte Carlo methods are entirely absent. Eight challenges are identified, including the complete absence of studies in tropical maritime, ultra-marine and coastal climates — motivating a future pilot DT for a bioclimatic building in such a context.

Keywords: Digital Twin, Smart Buildings, Energy Optimization, BIM, IoT, HVAC, PRISMA, systematic review.

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1. Introduction

Digital Twins (DTs) have rapidly emerged as one of the most transformative technological paradigms in the built environment [2]. Defined as dynamic digital replicas of physical assets continuously fed by real-time data, DTs enable advanced monitoring, predictive analytics, simulation, and automated control. In the context of Smart Buildings — complex environments integrating IoT devices, building automation systems, and intelligent control frameworks — Digital Twins promise a paradigm shift in the way buildings are designed, operated, and optimized. The increasing pressure to reduce energy consumption, enhance occupant comfort, and extend equipment lifespan has further accelerated the adoption of DT-based strategies, positioning Smart Buildings as a key application area for this technology.

Buildings are responsible for more than one-third of global final energy consumption, and improving their efficiency remains essential for meeting sustainability targets. Conventional building automation systems often rely on static rules and limited sensor integration, resulting in sub-optimal operational performance. Digital Twins, by contrast, provide a comprehensive and dynamic representation of building systems — such as HVAC, lighting, or energy storage — allowing real-time decision-making and predictive control. Recent advances in artificial intelligence, machine learning, and data-driven modeling have deepened the potential of DTs, enabling adaptive optimization strategies and accurate forecasting of energy demand, occupant behavior, and equipment degradation. Despite this rapid technological growth, the scientific landscape remains fragmented; studies differ significantly in their modeling methodologies, validation approaches, and degrees of real–virtual coupling.

This fragmentation motivates four explicit research questions which structure the present review:

- **RQ1** — In what application areas are Digital Twins currently applied to Smart Buildings, and how do these applications differ in terms of architecture, methodology, and degree of real–virtual coupling?
- **RQ2** — How effective are these implementations in improving energy performance, occupant comfort, or operating costs, and what key performance indicators (KPIs) are reported?
- **RQ3** — What challenges and technical limitations persist in this emerging field, and what research directions should be prioritized?
- **RQ4** — Which technical tools (programming languages, simulation engines, communication protocols) and methods (machine learning, control, simulation) are effectively mobilized by Digital Twins for buildings in the retained literature?

Addressing these questions is crucial for understanding both the maturity of existing DT technologies for building applications and the directions in which future research should evolve. This is particularly relevant for under-studied climatic contexts — tropical maritime, ultra-marine, and coastal environments — where the building sector faces specific cooling-dominated load profiles, high relative humidity, and salt-induced corrosion of HVAC equipment, conditions largely absent from the dominant European and East-Asian datasets.

In this context, a systematic review is essential to provide a rigorous and structured synthesis of the current state of the art. The PRISMA 2020 methodology [1] offers a transparent framework for identifying, screening, and selecting relevant scientific studies while ensuring reproducibility of the review process. By applying PRISMA — complemented by the systematic-review guidelines of Kitchenham and Charters [5] for evidence synthesis in software and systems engineering — this work consolidates the evidence on Digital Twin applications in Smart Buildings between 2015 and 2025, focusing specifically on studies that demonstrate validated implementations, real–virtual synchronization, and measurable impacts on energy performance or operational management. The review encompasses four major scholarly databases (Scopus, IEEE Xplore, Web of Science, and SpringerNature) and applies strict, principle-based inclusion and exclusion criteria so that only methodologically grounded contributions are analyzed.

The purpose of this introduction is therefore twofold: first, to position Digital Twins as a promising technological foundation for energy-efficient and intelligent building management; and second, to justify the need for a systematic, PRISMA-guided evaluation of the scientific evidence. By clarifying the motivations, research questions, and scope of the review, this section establishes the foundations for a comprehensive investigation into the role of Digital Twins in shaping the next generation of sustainable, intelligent buildings.

2. Methodology

This systematic review follows the PRISMA 2020 methodology [1] to ensure transparency, reproducibility, and scientific rigor. The review process consisted of four major phases — identification, screening, eligibility assessment, and inclusion — supported by a structured protocol, predefined inclusion and exclusion criteria, and a multi-stage screening pipeline involving both automated tooling and expert manual assessment. The taxonomic framework for classifying the included studies follows Kritzinger et al. [3] (Model / Shadow / Twin) and Madni et al. [4] (DT maturity 0–3).

2.1 Data Sources and Search Strategy

A comprehensive search was conducted across four major scientific databases widely used in engineering and building research:

- **Scopus**
- **IEEE Xplore**
- **Web of Science**
- **SpringerNature** (including ScienceDirect-indexed records)

To ensure broad coverage while maintaining thematic precision, carefully engineered Boolean search queries were applied consistently across all platforms.

Generic search query (used for IEEE Xplore, SpringerNature, and Web of Science):

```
("digital twin" OR "virtual twin" OR "digital replica" OR
"cyber-physical system")
AND ("smart building" OR "intelligent building" OR
"building energy management" OR "HVAC" OR "BIM" OR
"building information model*")
AND ("energy optimization" OR "energy efficiency" OR
"predictive control" OR "energy management" OR
"occupancy" OR "thermal comfort")
AND ("case study" OR "experimental" OR "simulation" OR
"validation")
NOT ("game" OR "metaverse" OR "twin flame")
```

Scopus-specific query (TITLE-ABS-KEY field):

```
TITLE-ABS-KEY( ("digital twin" OR "digital twins" OR
"virtual replica" OR "cyber-physical system")
AND ("smart building" OR "intelligent building" OR
"building information modeling" OR BIM OR
"energy management" OR HVAC)
AND ("energy optimization" OR "energy efficiency" OR
"predictive control" OR "thermal" OR "occupancy")
AND ("case study" OR "experimental" OR "simulation" OR
```

```
"validation")
AND NOT ("game" OR "metaverse" OR "twin flame"))
```

These queries intentionally target Digital Twin applications in Smart Buildings with a focus on experimental validation, energy-oriented performance, and real–virtual coupling, directly aligned with the scope and objectives of this review.

Search results. Searches were performed in October–November 2025 and returned the following raw record counts:

Database	Records retrieved
Scopus	343
Web of Science	150
SpringerNature	144
IEEE Xplore	67
Total (before cleaning)	704

All results were exported in CSV (or XLS for Web of Science) format, preserving metadata such as title, abstract, year, keywords, authors, and DOI.

2.2 Data Cleaning and Deduplication

All exported datasets underwent a rigorous, two-pass cleaning procedure implemented in Python (`pipeline_complet.py`).

Pass 1 — Standardization and intra-database deduplication. All datasets were normalized into a unified schema with the following columns: Title, Abstract, Year, Keywords (merged from author keywords and indexing terms), Source, DOI, and Authors. Duplicate records within each database were then removed using a hierarchical matching procedure: (i) first by normalized DOI (DOIs were lowercased and the `https://doi.org/` prefix stripped), and (ii) second by normalized title plus year for records with no DOI. Title normalization removed punctuation, applied Unicode NFKD normalization, and converted to lowercase before comparison.

Pass 2 — Inter-database deduplication. Records from the four databases were merged into a unified corpus and deduplicated again using the same two-step procedure (DOI first, then Title + Year). When duplicates were detected, the record with the longest abstract was retained, and an `_all_databases` field tracked the union of source databases. This step removed 156 inter-database duplicates.

After these two passes, **548 unique articles** constituted the unified corpus (`corpus_unified_screening_ready.csv`) that served as input to the screening phase.

2.3 Eligibility Criteria

Eligibility criteria were established *a priori* based on the objectives of the review. They were then reinforced during the strict re-sharpening phase (see §2.4.3).

Inclusion criteria (cumulative):

1. Study explicitly involving a Digital Twin (or Digital Shadow) applied to a building.
2. Demonstration of real–virtual synchronization (sensor data, real-time data flows, or simulation coupled with real building data).
3. Published between 2015 and 2025.
4. Experimental study, validated prototype, simulation with real-data coupling, or real-world case study.
5. Includes measurable outcomes related to energy performance, thermal comfort, HVAC optimization, or maintenance.
6. Article written in English.

Exclusion criteria:

1. Purely theoretical or conceptual papers with no empirical validation.
2. Smart-city–scale, district-scale, or neighborhood-scale studies not focused on individual buildings.
3. BIM-only or static simulations without real-data integration or real–virtual coupling.
4. Non-peer-reviewed materials (industry pitches, blogs, short editorials, posters without proceedings).
5. Duplicates or studies for which the full text could not be retrieved (these are nonetheless reported separately in the PRISMA flow diagram as “Reports not retrieved” in line with PRISMA 2020 guidelines).
6. Domains outside the building scope (manufacturing, vehicles, data centers, batteries, livestock, agriculture).

2.4 Multi-Stage Screening Process

To handle the volume of records while preserving methodological rigor, a multi-stage screening process was implemented. Each stage is documented transparently below; intermediate artefacts (CSV files, Python scripts, decision logs) are preserved in the project repository to ensure full reproducibility.

2.4.1 Stage 0 — Exploratory LLM-assisted pilot (Scopus subset) In an initial exploratory stage, an LLM-assisted screening pipeline was applied to the Scopus subset ($n = 147$ after intra-database deduplication on this single database). The Gemini 2.5 Flash model was prompted with the full PRISMA context, the inclusion and exclusion criteria, and a constrained JSON output schema (`{"decision": "In-clure"|"Exclure"|"Doute", "reason": "..."}).` This stage was used as a calibration exercise to assess the consistency of automated decisions and to identify ambiguous patterns. It did not directly produce the final included corpus.

The exploratory pilot revealed two limitations: (i) the original Scopus deduplication was incorrectly performed on the journal ISSN rather than the article DOI, leading to over-aggressive removal of valid articles; and (ii) the LLM was applied only to a single database, introducing a sampling bias. Both issues were subsequently corrected before the main screening (see §2.2 and §2.4.2).

2.4.2 Stage 1 — AI-assisted title + abstract screening (V2) The corrected unified corpus of **548** articles was screened with the assistance of a Large Language Model (Claude Opus, Anthropic). Each title and abstract was assessed against the inclusion and exclusion criteria; the LLM produced an initial decision (Inclure, Exclure, or Doute) with a written rationale. The decision was first reviewed by one team

member, then **subsequently re-checked and validated by all five members of the project team** before being recorded. Cases marked Doute were resolved through team discussion. Decisions and rationales are preserved in `screening_results_v2_corrected.csv`.

After this stage, **167** articles were retained as candidate inclusions, **374** were excluded, and 7 remained in Doute for the next stage.

2.4.3 Stage 2 — Strict re-sharpening (V3) The professor’s guidelines specify a target dataset of 40–80 papers. The 167 candidates retained at the end of Stage 1 exceeded this range, indicating that the Stage 1 screening was insufficiently selective. A strict re-sharpening was therefore applied through the introduction of four cumulative criteria (`screening_v3_strict.py`):

- **C1 — Explicit Digital Twin terminology.** The article must contain an explicit mention of “digital twin”, “digital twins”, “digital twinning”, “digital shadow”, or “digital shadows”. The terms “cyber-physical system” (CPS), “virtual twin”, or “digital replica” alone trigger a WEAK signal that requires further inspection.
- **C2 — Explicit validation language.** The abstract must contain an explicit validation marker (case study, experiment, prototype, testbed, pilot, deployment, field test, real-world, validation, demonstrator, benchmark, empirical). Softer markers such as evaluate, test, or monitor alone yield a WEAK signal.
- **C3 — Substantive abstract.** The abstract must contain at least 100 words to allow meaningful screening; abstracts of 50–99 words yield a WEAK signal; abstracts of fewer than 50 words trigger FAIL.
- **C4 — Individual-building scope.** The combination of title and abstract must not contain a strong urban/district scope marker (urban-scale, city-scale, district-scale, smart cities, neighborhood, energy community, “X households” with $X \geq 10$) unless an explicit individual-building anchor is also present (individual building, single building, office building, case study building, etc.).

Each article received a status of PASS, WEAK, or FAIL for each of the four criteria. Articles were then assigned to one of three buckets:

- **INCLUDE_AUTO** — all four criteria PASS;
- **EXCLUDE_AUTO** — at least one criterion FAIL;
- **BORDERLINE** — a mix of PASS and WEAK signals, requiring human arbitration.

This procedure produced **95 INCLUDE_AUTO**, **48 BORDERLINE**, and **24 EXCLUDE_AUTO**. The 48 borderline articles were then read individually (title + full abstract). An initial decision and written motive were produced by one team member, and **each borderline decision was subsequently reviewed and validated by all five members of the project team**, with disagreements resolved by team discussion. The final decisions are recorded in `screening_v3_borderline_decisions.csv`. After this peer-validated arbitration, **35** of the 48 borderline articles were included and **13** were excluded.

The combined output of Stage 2 is **130 articles retained** for the eligibility (full-text) phase, **37 articles excluded** with documented reasons.

2.4.4 Stage 3 — Full-text retrieval and eligibility screening For each of the 130 retained articles, full-text retrieval was attempted using a tiered strategy combining the Unpaywall API (DOI-based), the OpenAlex API, the Semantic Scholar API, and the CORE.ac.uk API. For articles inaccessible through open-access aggregators, full-text was retrieved manually through institutional access (Université de Lille / Lillocat).

Of the 130 candidates, **51** full-text documents were successfully retrieved. The remaining **79** could not be obtained, mostly due to IEEE conference paywall restrictions or absence of an open-access version reachable from the available institutional licenses. These 79 articles are reported separately in the PRISMA flow diagram as “Reports not retrieved”, in line with the PRISMA 2020 guidelines.

The 51 retrieved full-texts were then read and assessed with LLM (Claude Opus, Anthropic) assistance under human supervision. Each article was systematically processed using a structured rapid-reading procedure (10–15 min per article) examining: abstract, introduction, methodology section (focusing on the DT architecture), validation/results section, and limitations. Each article received a final decision recorded with the following metadata: `fulltext_decision`, `kritzing_level` (Model / Shadow / Twin), `building_scope`, `validation_type`, `domain`, and a concise motif. **An initial decision was produced by one team member, and each full-text decision — both inclusions and exclusions — was subsequently reviewed and validated by all five members of the project team.** After this assessment, **45** articles met all the inclusion criteria and were retained for the qualitative synthesis. **6** articles were excluded after full-text reading, with the following reasons:

- 1 article was a literature review without a primary contribution;
- 2 articles applied DT to district or multi-region scales not visible from the abstract;
- 1 article addressed mega-facilities (hospitals + airports complexes) inconsistent with the individual-building scope;
- 2 articles described a “DT vision” or preparatory infrastructure rather than an implemented Digital Twin (Kritzing Model level only).

2.5 PRISMA 2020 Flow Diagram

The complete identification and screening process is summarized in Figure 1.

PRISMA 2020 Flow Diagram

Digital Twins for Energy Optimization in Smart Buildings — Systematic Review

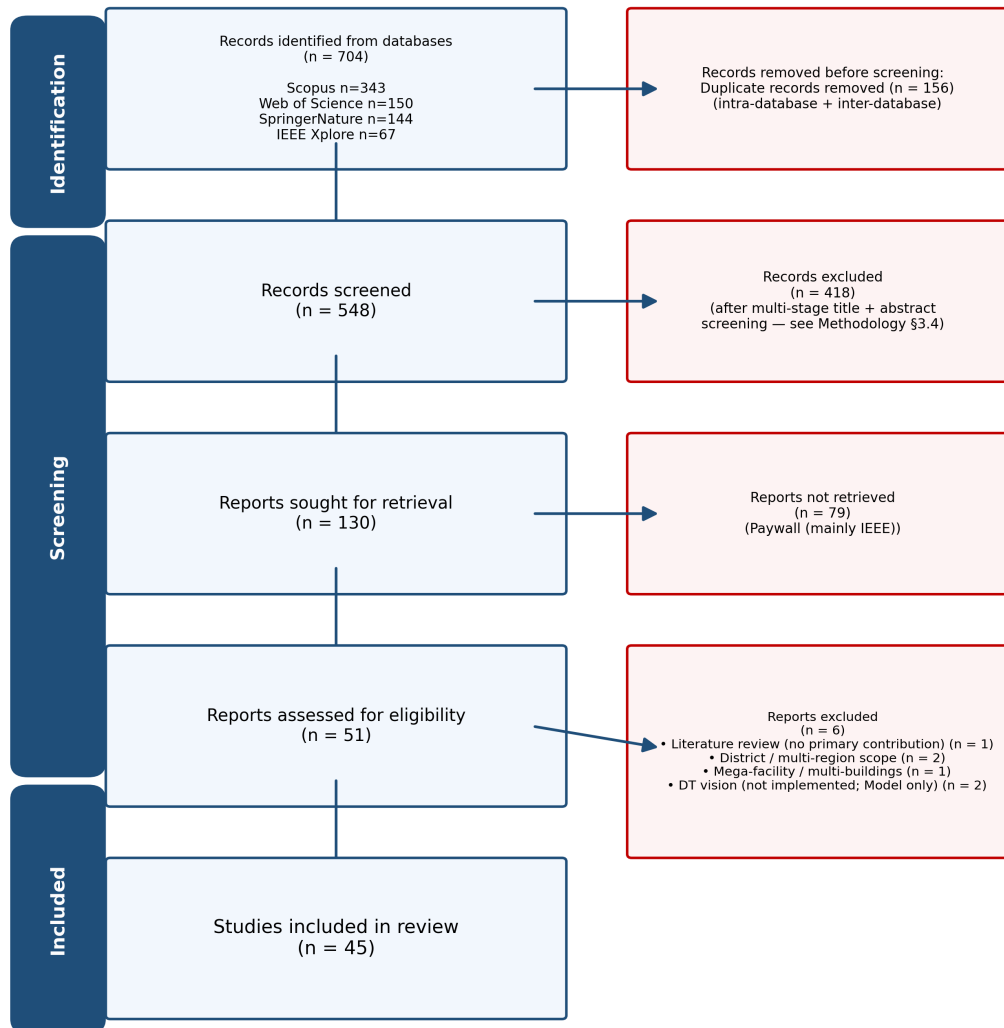
Adapted from Page MJ et al. The PRISMA 2020 statement. *BMJ* 2021;372:n71. <https://doi.org/10.1136/bmj.n71>

Figure 1 — Multi-stage screening process from 704 initial records to 45 included studies. The diagram synthesizes the procedure detailed in §2.4. The “Records excluded” box (n = 418) aggregates the Stage 1 AI-assisted title + abstract screening (374 excluded + 7 doute reclassified as exclusions) and the Stage 2 strict re-sharpening (24 EXCLUDE_AUTO + 13 borderline excluded after human arbitration). The “Reports not retrieved” box (n = 79) corresponds to articles for which the full text could not be obtained through

institutional access, mainly because of paywall restrictions on IEEE conference proceedings. Following the PRISMA 2020 standard [1].

2.6 Technical Mapping (RQ4)

To support RQ4, an additional pass over the 45 retained PDFs was performed to extract the technical stack (`extract_technical_details.py`). For each PDF, the text of the first five pages and the last three pages was scanned with a curated set of 60+ regular expressions (case-insensitive) covering: programming languages, simulation engines, BIM tools, communication protocols, IoT components, ML methods, simulation types, and validation modes. Detection is binary at article level (1 if at least one occurrence is matched, 0 otherwise). This mapping is descriptive, not inferential: a *mention* of a tool is not strictly equivalent to its operational use, and the regex coverage may miss artefacts due to PDF text-extraction noise. These limitations are discussed in §4.4.

3. Findings

3.1 Bibliometric Overview

Among the 45 retained studies, **78 % were published between 2023 and 2025**, confirming the rapid recent maturation of the Digital Twin field for Smart Buildings. The yearly distribution (Table 1) shows an exponential growth curve: only 2 articles per year on average between 2017 and 2020 (10 articles total over 4 years), 13 in 2023, 13 in 2024, and 9 in the first half of 2025 alone.

Table 1 — Distribution of included studies by year

Year	n	%
2017	1	2.2
2018	1	2.2
2019	2	4.4
2020	2	4.4
2021	3	6.7
2022	1	2.2
2023	13	28.9
2024	13	28.9
2025	9	20.0

Geographically, 49 % of the studies were conducted in four countries (Italy, USA, China, Spain). No included study was conducted in a tropical maritime, ultra-marine, or coastal context, and only two studies were conducted in arid contexts (UAE, Egypt). This geographic concentration is discussed further in §4.

According to the Kritzinger taxonomy [3], 71 % of the included studies (32/45) implemented a true Twin (bidirectional automatic flow between physical and virtual instances), while 29 % (13/45) remained at the Shadow level (unidirectional flow from physical to virtual without automated feedback). No included study was classified as a Digital Model only (i.e., without real-data coupling); such studies were filtered out at the screening stage.

The validation strategies of the included studies are summarized in Table 2. A notable observation is that **40 % of the studies relied solely on simulation**, without experimental or pilot validation in a physical building, a limitation discussed in §4.

Table 2 — Validation type of the included studies

Validation type	n	%
Simulation only	18	40.0
Pilot (single case study, real building)	13	28.9
Real deployment (ongoing field operation)	11	24.4
Testbed / Hardware-in-the-Loop	3	6.7

3.2 Thematic classification

The 45 included studies were classified into four mutually exclusive thematic categories based on their primary application area (Table 3). Each study was assigned to exactly one category. This classification structures the per-category analyses presented in §3.4 to §3.7. The previous dual split between energy management and HVAC control was consolidated into a single “Energy consumption & supervision” category, since the underlying methods (MPC, energy-signature regression, BIM/IoT integration) and the reported KPIs (energy savings, indoor temperature error) are essentially shared across the two sub-domains.

Table 3 — Thematic distribution of the 45 included studies

Category	n	%
(1) Energy consumption & supervision	32	71.1
(2) Indoor Comfort & Indoor Air Quality (IAQ)	9	20.0
(3) Maintenance & Structural Health	3	6.7
(4) Specialized Applications	1	2.2

3.3 Per-article taxonomy (RQ1)

Table 4 maps each of the 45 included studies to its category, year, Kritzinger level, and validation type. Reference numbers follow the IEEE bibliography of §6. Internal corpus identifiers are kept in column “ID” for traceability with the screening artefacts (included_papers_v4_classified.csv).

Table 4 — Per-article classification (RQ1)

Ref	ID	Year	Category	Kritzinger	Validation
[8]	24	2023	C1 — Energy consumption & supervision	Twin	Real deployment
[9]	261	2025	C1 — Energy consumption & supervision	Twin	Real deployment
[10]	112	2023	C1 — Energy consumption & supervision	Shadow	Real deployment
[11]	89	2024	C1 — Energy consumption & supervision	Twin	Pilot
[12]	547	2020	C1 — Energy consumption & supervision	Shadow	Simulation only
[13]	250	2024	C1 — Energy consumption & supervision	Twin	Simulation only
[14]	114	2025	C1 — Energy consumption & supervision	Twin	Simulation only
[15]	139	2024	C1 — Energy consumption & supervision	Twin	Simulation only

Ref	ID	Year	Category	Kritzinger	Validation
[16]	94	2023	C1 — Energy consumption & supervision	Twin	Simulation only
[17]	54	2023	C1 — Energy consumption & supervision	Twin	Simulation only
[18]	78	2017	C1 — Energy consumption & supervision	Twin	Real deployment
[19]	142	2025	C1 — Energy consumption & supervision	Twin	Real deployment
[20]	199	2025	C1 — Energy consumption & supervision	Twin	Simulation only
[21]	72	2023	C1 — Energy consumption & supervision	Twin	Simulation only
[22]	150	2023	C2 — Indoor Comfort & IAQ	Twin	Pilot
[23]	53	2025	C2 — Indoor Comfort & IAQ	Twin	Simulation only
[24]	226	2024	C2 — Indoor Comfort & IAQ	Twin	Pilot
[25]	74	2024	C3 — Maintenance & Structural Health	Twin	Simulation only
[26]	121	2023	C3 — Maintenance & Structural Health	Twin	Pilot
[27]	235	2021	C3 — Maintenance & Structural Health	Shadow	Pilot
[28]	306	2023	C4 — Specialized Applications	Twin	Pilot
[29]	195	2023	C1 — Energy consumption & supervision	Shadow	Real deployment
[30]	153	2024	C1 — Energy consumption & supervision	Twin	Real deployment
[31]	106	2019	C1 — Energy consumption & supervision	Shadow	Simulation only
[32]	258	2024	C1 — Energy consumption & supervision	Twin	Pilot
[33]	327	2018	C1 — Energy consumption & supervision	Twin	Testbed
[34]	25	2024	C1 — Energy consumption & supervision	Twin	Testbed
[35]	6	2024	C1 — Energy consumption & supervision	Shadow	Simulation only
[36]	33	2023	C1 — Energy consumption & supervision	Twin	Pilot
[37]	42	2024	C2 — Indoor Comfort & IAQ	Twin	Real deployment
[38]	67	2021	C1 — Energy consumption & supervision	Shadow	Pilot
[39]	88	2025	C1 — Energy consumption & supervision	Shadow	Pilot

Ref	ID	Year	Category	Kritzinger	Validation
[40]	99	2023	C1 — Energy consumption & supervision	Twin	Simulation only
[41]	124	2025	C1 — Energy consumption & supervision	Twin	Real deployment
[42]	132	2019	C1 — Energy consumption & supervision	Shadow	Real deployment
[43]	166	2023	C1 — Energy consumption & supervision	Shadow	Simulation only
[44]	170	2025	C2 — Indoor Comfort & IAQ	Twin	Simulation only
[45]	189	2025	C2 — Indoor Comfort & IAQ	Shadow	Pilot
[46]	193	2025	C2 — Indoor Comfort & IAQ	Twin	Pilot
[47]	270	2025	C2 — Indoor Comfort & IAQ	Twin	Simulation only
[48]	272	2023	C2 — Indoor Comfort & IAQ	Shadow	Simulation only
[49]	307	2023	C1 — Energy consumption & supervision	Twin	Simulation only
[50]	341	2020	C1 — Energy consumption & supervision	Twin	Pilot
[51]	366	2021	C1 — Energy consumption & supervision	Twin	Testbed
[52]	535	2022	C1 — Energy consumption & supervision	Shadow	Real deployment

Aggregated counts: C1 = 32, C2 = 9, C3 = 3, C4 = 1; Twin = 32, Shadow = 13; Simulation only = 18, Pilot = 13, Real deployment = 11, Testbed = 3 (consistent with Tables 2 and 3).

3.4 Category 1 — Energy consumption & supervision (n = 32)

This category groups studies whose primary objective is the optimization of overall building energy consumption, performance, and HVAC operation. It absorbs both the energy-management/performance-prediction and the HVAC/thermal-control sub-domains, since the underlying modelling stacks (BIM + IoT + ML, MPC, energy-signature regression) and the reported KPIs (energy savings %, indoor temperature error, MPC cost) are largely shared. It is by far the largest category (71 % of the corpus).

Representative studies — energy management and performance prediction. Hadjidemetriou et al. [8] propose a multi-domain DT architecture combining ordinary differential equations and computational fluid dynamics, validated in a real building with an average zone temperature error below 1 %. Bjørnskov et al. [9] demonstrate an ontology-driven DT framework on a hospital floor with 12 zones and 45 sensors, achieving a mean absolute error of 0.40 °C for temperature and 32 ppm for CO₂. Manfren et al. [10] introduce a “lean and interpretable” DT based on energy-signature regression with a rolling horizon, deployed in real buildings to verify smart thermostatic radiator valve (TRV) and gas absorption heat pump retrofits.

Representative studies — HVAC and thermal control. Schmidt et al. [18] — the earliest included study

— deployed a cyber-physical system framework on a Spanish school over 43 winter days in 2015–2016, achieving energy savings of approximately one third while maintaining thermal comfort. Lee et al. [19] propose an Adaptive Domain Nelder–Mead calibration method for HVAC DTs, reducing temperature RMSE by 60–70 % compared with the traditional Nelder–Mead algorithm. Andersen and Jradi [20] implement a multi-objective Model Predictive Control (MPC) DT on an 8 500 m² university building, balancing electricity cost, indoor air quality, and CO₂ emissions, achieving 18 % energy savings versus a rule-based baseline.

Methodological trends. Three sub-approaches dominate this category. (i) Pure data-driven approaches (e.g., Han et al. [11], Kathirgamanathan et al. [12], Stogia et al. [13]) leverage machine learning for consumption forecasting with limited physical coupling; reported energy savings range from 18 % to 30 %. (ii) Physics-based or white-box approaches (e.g., Ramirez et al. [14], Engel et al. [15], Lv et al. [16]) rely on EnergyPlus, Modelica, or TRNSYS simulations calibrated against measured data, providing higher modelling fidelity at increased computational cost. (iii) Hybrid approaches (e.g., Bjørnskov et al. [9], Manfren et al. [10], Song et al. [17]) combine semantic ontologies or grey-box models with data-driven calibration and demonstrate superior generalizability. Within HVAC studies specifically, Model Predictive Control is the dominant control strategy, followed by data-driven prediction and CPS-hybrid architectures. The reported median energy savings across the category is approximately 20–25 %, with a range of 15–33 %.

Validation rigor. Of the 32 studies in this category, 10 report real-deployment validation, 6 are pilot studies, 13 remain at the simulation-only stage, and 3 use a testbed / Hardware-in-the-Loop setup. The intra-category split (Table 4) shows that the HVAC sub-domain exhibits the highest ratio of real-deployment validation, reflecting the relative measurability of HVAC outcomes compared with broader energy management.

Heritage buildings. Two studies (Zhang et al. [21]; Schmidt et al. [18]) target heritage buildings, an emerging niche where DT-driven low-energy IAQ control has shown energy savings of up to 30 % while preserving structural integrity.

3.5 Category 2 — Indoor Comfort & Indoor Air Quality (n = 9)

This category concentrates on thermal-comfort prediction, occupant-centric models, and indoor-air-quality monitoring.

Representative studies. Edtmayer et al. [22] propose a coupled Building Energy Simulation (BES) and Computational Fluid Dynamics (CFD) DT for virtual sensors in a test facility, achieving 0.35 K mean absolute error for indoor temperature and 1.2 % for relative humidity over five consecutive days, with a total computation time below 10 minutes (compared with 20 hours for the corresponding pure CFD model). Ma et al. [23] introduce **Co-zyBench**, a co-simulation benchmarking tool that uses one DT for the building and HVAC system and another for the occupants, allowing standardized evaluation of personalized thermal-comfort provisioning systems. De Rubeis et al. [24] document a complete BIM-based thermo-hygrometric DT workflow applied to an office building used as an academic Living Lab.

Methodological trends. This category is distinctively BIM- and IoT-centric: 6/9 studies build their DT on a BIM model coupled with IoT sensor streams. Personalized comfort models (e.g., simplified PMV variants, hybrid Prophet + LSTM, ANN surrogates) emerge as a clear trend across 3 studies. Validation is balanced between pilot (4/9) and simulation-only (4/9), with a single real-deployment ([37]); no testbed setup was

found in this category, which constitutes a critical observation for RQ3 since IAQ outcomes are sensitive to occupancy patterns that are difficult to reproduce in simulation.

3.6 Category 3 — Maintenance & Structural Health (n = 3)

This small category groups studies on predictive maintenance, fault detection, and structural conservation.

Cheng Zhang et al. [25] couple a Heritage Building Information Modeling (HBIM) DT with CFD and structural finite-element simulations to control heater placement and operating settings, reducing maximum total deformation of a heritage building by 62.9 %. Hosamo et al. [26] integrate BIM, real-time sensor data, occupant feedback, and Bayesian Networks into a DT framework that detects and predicts HVAC faults affecting comfort; the framework was validated on two Norwegian buildings with data spanning 2019–2022. Seghezzi et al. [27] describe an occupancy-oriented DT framework based on IoT camera-based sensors, with calibration and test campaigns at the Politecnico di Milano.

Observation. The under-representation of predictive maintenance applications (only 7 % of the included corpus) is a clear challenge in the literature, especially given the maturity of predictive-maintenance approaches in adjacent industries such as manufacturing.

3.7 Category 4 — Specialized Applications (n = 1)

Cairolì and Tagliabue [28] describe a Digital Twin for acoustics and stagecraft facility management in a multipurpose hall (the Roberto De Silva Hall), where the DT controls reverberation time in real time as a function of stage equipment layouts, variable acoustic elements, and audience occupancy. The isolated nature of this study confirms that DT applications are beginning to extend beyond the classical energy/comfort axis toward more specialized building functions, but that such niches remain emerging.

3.8 Effectiveness KPIs reported (RQ2)

Reported KPIs are heterogeneous in nature and granularity, which prevents a strict numerical meta-analysis. Table 5 summarizes the dominant KPI families per category, with representative quantitative outcomes and their article references. Cross-category extremes are kept in the prose rather than the table to avoid spurious comparison.

Table 5 — Effectiveness KPIs reported per category (RQ2)

Category	KPI family	Representative outcomes (article ref)
C1 — Energy consumption & supervision	Energy savings (%)	18–30 % data-driven [11], [12], [13]; ≈ 33 % CPS [18]; 18 % MPC [20]; 30 % heritage [21]
	Indoor temperature error	< 1 % [8]; 0.40 °C MAE [9]; RMSE –60–70 % vs Nelder–Mead [19]; RMSE 0.16 °C [29]

Category	KPI family	Representative outcomes (article ref)
C2 — Indoor Comfort & IAQ	MPC cost / OPEX	530 MWh/year, 641 t CO ₂ /year [32]; ROI 23 yr NZEB [31]
	Co-simulation runtime	10 min vs 20 h pure CFD [22]
	Comfort metrics (PMV, IEQ)	0.40 °C / 32 ppm CO ₂ [9]; 0.35 K MAE / 1.2 % RH [22]; PMV / sPMV variants [37], [44], [47]
C3 — Maintenance & Structural Health	Personalized comfort	Co-zyBench benchmark [23]; Prophet + LSTM hybrids [37]
	Fault detection rate	Validated on 2 buildings, 2019–2022 [26]
	Structural metrics	–62.9 % max deformation [25]
	Sensor calibration	Test campaigns Politecnico di Milano [27]
C4 — Specialized Applications	Reverberation control	Real-time RT60 control [28]

Cross-category quantitative summary. Reported energy savings across the corpus span 15–33 %, with a median of approximately 20–25 %. Predictive accuracies range from RMSE 0.16 °C [29] to MAE 0.045 [30], again with heterogeneous baselines. Cost-related outcomes are sparse — only [31] and [32] report explicit ROI or absolute savings, and none provides a normalized cost-per-square-metre.

3.9 Cross-cutting trends

Beyond the per-category analyses above, several trends cut across the entire corpus.

Methodological stack. Forty percent of the included studies (18/45) combine BIM, IoT sensors, and machine learning in a hybrid stack — the dominant pattern in the field. Pure physics-based modelling (EnergyPlus, TRNSYS, Modelica) is used in 24 % of studies (11/45), and pure data-driven approaches in 20 % (9/45). MPC-based control appears in 13 % (6/45), CPS architectures explicitly invoked in 11 % (5/45), CFD coupling in 9 % (4/45), and reinforcement learning in only 4 % (2/45).

Communication and platforms. BIM environments (Revit, IFC, Dynamo) are mentioned in 49 % of the studies; IoT sensors (Wi-Fi probe, temperature, humidity, CO₂, occupancy PIR, energy meters) in 53 %; cloud or edge platforms (Azure DT, AWS, custom academic prototypes) in 24 %. The MQTT communication protocol is explicitly mentioned in only 11 % of the studies, even though many architectures rely on similar publish–subscribe mechanisms — suggesting that protocol-level details are systematically under-documented in the reviewed literature.

Quantitative outcomes. Across all categories, the reported energy savings span from 15 % to 33 %, with a median of approximately 20–25 %. Comfort-related outcomes are reported with heterogeneous KPIs (PMV, sPMV, IEQ score, occupant satisfaction surveys), and the absence of standardized metrics prevents formal numerical meta-analysis.

3.10 Technical landscape (RQ4)

To answer RQ4, the 45 retained PDFs were scanned for the presence of programming languages, simulation engines, communication protocols, ML methods, and validation modes (cf. §2.6). Table 7 reports tool / framework prevalence; Table 8 reports method prevalence. Each row lists every article in which the corresponding pattern was detected. Detection is binary at article level and reflects *mention*, not strictly verified *operational use*.

Table 7 — Tools and frameworks (RQ4-A)

Family	Tool / standard	n	Articles (IEEE refs)
Building Energy Simulation	EnergyPlus	15	[8], [9], [12], [14], [19], [20], [23], [34], [36], [41], [42], [43], [44], [48], [49]
	Modelica / Dymola	10	[9], [15], [19], [20], [23], [34], [36], [42], [50], [51]
	TRNSYS	7	[8], [19], [36], [39], [41], [48], [49]
	IDA-ICE / OpenStudio	5	[21], [22], [36], [39], [41]
BIM / CAD	IFC standard	9	[9], [18], [21], [26], [37], [39], [45], [46], [49]
	Revit	7	[21], [25], [26], [31], [37], [38], [45]
	Dynamo	4	[26], [28], [31], [38]
Co-simulation	FMI / FMU	3	[30], [34], [51]
	Co-simulation (generic)	4	[22], [23], [30], [48]
Languages	Python (+ ML stack)	12	[9], [10], [12], [15], [26], [28], [37], [39], [41], [44], [49], [51]
	MATLAB / Simulink	5	[8], [16], [17], [22], [48]
Communication	HTTP / REST API	8	[8], [16], [23], [34], [36], [37], [43], [45]
	MQTT	5	[13], [14], [46], [50], [51]
	BACnet / Modbus	7	[18], [21], [26], [34], [41], [44], [46]

Family	Tool / standard	n	Articles (IEEE refs)
IoT layer	OPC-UA	0	(none)
	Generic IoT mention	29	broad — see corpus extraction CSV for full list

Reading. EnergyPlus dominates the simulation layer; Python is the de facto language; the communication layer is overwhelmingly web-oriented (HTTP/REST, MQTT) rather than industrial (BACnet, OPC-UA), signalling a research/IT positioning rather than a field-engineering one. The complete absence of OPC-UA in the corpus is notable and is discussed in §4.

Table 8 — Methods (RQ4-B)

Family	Method	n	Articles (IEEE refs)
Predictive control	MPC	22	[9], [10], [12], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [29], [34], [35], [36], [39], [44], [46], [47], [50]
Deep learning	ANN / MLP	13	[15], [16], [18], [23], [26], [29], [30], [36], [39], [44], [45], [47], [48]
	LSTM	9	[9], [14], [15], [17], [19], [30], [37], [47], [48]
Reinforcement learning	CNN	4	[17], [30], [32], [36]
	RL / DQN / Q-learning	9	[15], [17], [18], [19], [23], [30], [34], [39], [44]
Classical (deterministic)	Random Forest / XGBoost	7	[12], [15], [26], [29], [42], [44], [47]
	Bayesian / Gaussian Process	4	[19], [23], [26], [50]
Stochastic / uncertainty			
Metaheuristics	GA / NSGA	9	[8], [14], [18], [19], [26], [30], [39], [41], [44]
	PSO	8	[18], [19], [29], [30], [35], [36], [41], [44]

Family	Method	n	Articles (IEEE refs)
Specialised simulation	CFD	5	[8], [21], [22], [25], [37]
Validation evidence	Co-simulation	4	[22], [23], [30], [48]
	Hardware-in-the-Loop	5	[8], [22], [30], [34], [51]
	Testbed	6	[12], [22], [26], [34], [36], [48]

Reading. Model Predictive Control is the pivot method (≈ 1 article in 2). Deep learning concentrates on ANN and LSTM; reinforcement learning, while minority (9/45), is conceptually central since the DT typically serves as the simulated environment in which RL is trained — a use case that is emblematic of the DT paradigm. Bayesian / GP and Monte Carlo methods (the latter detected in 0 / 45 articles) are striking by their scarcity, signalling a limited probabilistic maturity of the DT-building literature.

4. Discussion

4.1 Answers to the research questions

RQ1 — Application areas of DT in Smart Buildings. The four categories identified in §3.2 reflect a clear hierarchy of maturity. Energy consumption & supervision is by far the most mature area (71 % of the corpus), benefiting from the early convergence of BIM, IoT, machine learning, and Model Predictive Control across both energy management and HVAC sub-domains. Indoor comfort and IAQ (20 %) is a fast-growing area in which BIM–IoT integration dominates. Maintenance and structural health (7 %) and specialized applications (2 %) remain emerging niches. The architectural dichotomy revealed by the Kritzinger classification — 71 % Twin versus 29 % Shadow — confirms that the field has progressed beyond pure monitoring toward closed-loop bidirectional implementations, although a non-trivial fraction of studies still publish “DT” frameworks that, on close reading, implement only a Shadow.

RQ2 — Effectiveness in energy, comfort, and operating-cost performance. Reported energy savings concentrate around 20–25 % (range 15–33 %), with comfort improvements quantified through heterogeneous KPIs (PMV deltas, IEQ scores, satisfaction percentages). Cost-related outcomes are sparser: only a few studies report explicit ROI (e.g., 23 years for an NZEB retrofit in Kaewunruen et al. [31]) or absolute savings (e.g., 530 MWh/year and 641 t CO₂/year in Piras et al. [32]). The heterogeneity of baselines — rule-based control, regression, MPC, traditional Nelder–Mead — and of physical contexts prevents a formal numerical meta-analysis. We therefore consciously chose a structured narrative synthesis for §3 rather than a pseudo-aggregate mean. Climatic context appears as a second-order modulator that is rarely reported and almost never analyzed as such; this challenge is critical for translation to non-temperate contexts.

RQ3 — Challenges and technical limitations. Eight challenges and technical limitations were identified across the corpus. They are listed in Table 6 with article exemplars (or the absence thereof, where relevant).

Table 6 — Challenges and technical limitations identified (RQ3)

#	Challenge / limitation	Quantitative evidence	Article exemplars
1	Over-reliance on simulation-only validation	18 / 45 = 40 %	sim-only: [12], [13], [14], [15], [16], [17], [20], [21], [23], [25], [31], [35], [40], [43], [44], [47], [48], [49]
2	Geographic concentration in temperate climates	49 % in 4 countries; 0 % tropical maritime	bulk of corpus
3	Under-representation of predictive maintenance	3 / 45 = 7 %	only [25], [26], [27]
4	No standardized DT platform	Azure DT / Eclipse Ditto / custom prototypes coexist	platform diversity throughout corpus

#	Challenge / limitation	Quantitative evidence	Article exemplars
5	Communication protocols under-documented	MQTT 11 %, OPC-UA 0 %, BACnet/Modbus 16 %	Table 7
6	Limited cross-building generalization	majority single-case studies	bulk of corpus
7	Personalized comfort models rarely validated in real settings	C2: only 1 / 9 real-deployment	[22], [23], [24], [44], [45], [46], [47], [48]
8	Cyber-security largely absent	1 / 45 explicit cyber-security mention	only [34]

RQ4 — Tools and methods effectively mobilized. Tables 7 and 8 in §3.10 show that EnergyPlus, Python, and Model Predictive Control form the dominant technical core of the field. The communication layer is overwhelmingly web-oriented (HTTP/REST, MQTT) rather than industrial (BACnet, OPC-UA). Reinforcement learning, while present in only 9 / 45 studies, is conceptually central since the DT typically serves as the simulated environment in which RL is trained — a use case that is emblematic of the DT paradigm. Probabilistic methods (Monte Carlo, Bayesian inference, Gaussian processes) are underused: Monte Carlo is detected in 0 / 45 articles and Bayesian / GP in only 4 / 45, signalling a limited probabilistic maturity of the DT-building literature.

4.2 Methodological maturity of the field

The temporal distribution (Table 1) shows that 78 % of the included studies were published in the last three years (2023–2025). The field is therefore still in a phase of rapid expansion, characterized by methodological diversity rather than convergence. The dominance of BIM + IoT + ML hybrid stacks (40 %) suggests that this combination is becoming a *de facto* standard, but the persistent diversity of underlying tools (EnergyPlus vs Modelica vs TRNSYS vs custom; Azure DT vs Eclipse Ditto vs prototype) indicates that no platform consensus has yet emerged. The Kritzinger split (71 % Twin / 29 % Shadow) suggests that researchers increasingly recognize the importance of bidirectional coupling, but a substantial minority of studies still describe a Shadow-level system as a “Digital Twin”, which contributes to terminological dilution in the field.

4.3 Strengths and weaknesses of current approaches

The strengths of current DT approaches in Smart Buildings are clear. The integration of real-time IoT data with BIM models enables granular monitoring; machine-learning components support occupant-aware adaptation; and Model Predictive Control coupled with physics-based DTs produces controllable energy savings of 20–25 %. The use of ontologies (e.g., Bjørnskov et al. [9]) and lean interpretable approaches (Manfren et al. [10]) further suggests that a second generation of DT, focused on interoperability and explainability, is now emerging.

The weaknesses are equally clear. First, the dominance of simulation-only validation (40 %) limits the external validity of reported energy savings. Second, the absence of standard benchmarks (with the partial exception of Co-zyBench and BAS Benchmarks; Ma et al. [23]; Cauchi & Abate [33]) makes cross-study comparison difficult. Third, the geographic concentration in temperate Europe and East Asia means that the literature is largely untested in tropical, arid, or maritime climates, where cooling-dominated load profiles, humidity, and corrosion behave very differently. Fourth, the under-representation of predictive maintenance and the near-absence of cyber-security indicate that the operational realities of building-management practice are still imperfectly addressed by the academic literature.

4.4 Limitations of this review

This review is subject to five limitations that the reader should keep in mind.

1. **Reports not retrieved (n = 79).** Of the 130 articles retained at the end of the title + abstract screening, 79 could not be retrieved as full text, mostly because of paywall restrictions on IEEE conference proceedings unavailable through the Université de Lille institutional access. This may introduce a slight bias toward open-access editors (MDPI, Frontiers, IEEE Open Access, Elsevier ScienceDirect with institutional subscription) at the expense of paywalled IEEE conference papers. We mitigated this bias by documenting all 79 unretrieved articles transparently in the PRISMA flow diagram and by listing them in the project repository for any future replication.
2. **Single-LLM exploratory pilot.** Stage 0 of the screening used a single Large Language Model (Gemini 2.5 Flash) on the Scopus subset only. This pilot was used as a calibration exercise and did not directly produce the final corpus.
3. **Peer validation across the team without independent double-screening.** All manual screening decisions — Stage 1 LLM-assisted title and abstract assessment, Stage 2 V3 borderline arbitration, and Stage 3 full-text reading — were initially produced by one team member and subsequently reviewed and validated by all five members of the project team, with disagreements resolved by team discussion. While stronger than a single-reviewer protocol, this peer-validation procedure is not strictly equivalent to a PRISMA-recommended double-reviewer protocol, in which each article is screened independently by two reviewers and Cohen's kappa is then computed on the disagreements. A formal independent double-screening with kappa computation is left for future work.
4. **No formal numerical meta-analysis.** Because of the heterogeneity of KPIs, baselines, and physical contexts, we deliberately chose a structured narrative synthesis rather than a pseudo-aggregate quantitative meta-analysis. We consider this the more rigorous choice in the present state of the field.
5. **AI-assisted screening, full-text reading, and technical extraction.** All stages of the screening pipeline (Stage 0 exploratory pilot via Gemini 2.5 Flash; Stage 1 title + abstract screening and Stage 3 full-text reading via Claude Opus, Anthropic) used Large Language Model assistance under human supervision. The technical extraction supporting RQ4 (§2.6, §3.10) is regex-based and reflects *mention* of a tool or method, not strictly verified *operational use*; PDF text-extraction noise may also produce minor false negatives. All borderline decisions and final inclusion outcomes were reviewed and validated by the full five-member project team (see limitation 3 above for the precise scope of

this peer-validation procedure); intermediate artefacts (CSV files, decision logs, Python scripts) are preserved for full reproducibility. While AI-assisted PRISMA reviews are increasingly accepted [6], [7], this approach is declared transparently here. A formal Cohen's kappa between human and LLM decisions on a 20 % sample was not computed and is left for future work.

4.5 Implications for tropical, ultra-marine, and coastal contexts

None of the 45 included studies was conducted in a tropical maritime, ultra-marine, or coastal climate. This is a striking challenge with direct implications for a substantial fraction of the global building stock that operates under conditions fundamentally different from those represented in the literature.

Three structural particularities distinguish these contexts from the dominant European and East-Asian datasets and call for dedicated empirical investigation:

1. **Cooling-dominated load profile.** In tropical climates, the annual energy balance of buildings is inverted compared with temperate latitudes: HVAC loads are dominated by cooling and dehumidification rather than heating. Control strategies, predictive models, and energy KPIs validated in heating-dominated contexts may not transpose directly.
2. **High and persistent humidity.** Indoor air quality control, condensation management, and material degradation behave differently under sustained humidity above 70 %. This affects both the IAQ-oriented Digital Twins of Category 2 and the predictive-maintenance frameworks of Category 3.
3. **Salt-induced corrosion and equipment degradation in coastal / ultra-marine settings.** Salt aerosols accelerate the wear of HVAC heat exchangers, electronics, and sensor enclosures. The few maintenance-oriented Digital Twins of the corpus do not address this failure mode.

A direct transposition of the reported 20–25 % energy-saving figures to tropical-maritime or coastal conditions cannot be assumed without further empirical validation. Future work should specifically test the most methodologically rigorous frameworks of §3.4–§3.7 (Hadjidemetriou et al. [8]; Bjørnskov et al. [9]; Andersen and Jradi [20]) on pilot buildings located in tropical maritime or ultra-marine territories, with particular attention to humidity dynamics and corrosion-aware predictive maintenance. Such studies would directly address a clear blank space in the international literature and contribute primary evidence to a research community that currently has very little of it.

5. Conclusion

This systematic review applied the PRISMA 2020 methodology [1] to consolidate the evidence on Digital Twin applications in Smart Buildings between 2015 and 2025. Following a four-database search yielding 704 records, a multi-stage screening process — combining a deduplicated unified corpus, expert manual title and abstract assessment, a strict re-sharpening with four cumulative criteria, and full-text reading — produced a final corpus of 45 included studies, in line with the recommended size for a Master-level systematic review.

Four principal findings emerge. First, the field is in a phase of rapid maturation, with 78 % of the included studies published between 2023 and 2025; the dominant methodological stack combines BIM, IoT, and machine learning, and the technical core converges on EnergyPlus, Python, and Model Predictive Control. Second, reported energy savings concentrate around 20–25 %, with the HVAC sub-domain of Category 1 showing the most rigorous validation while the broader energy-management sub-domain relies disproportionately on simulation-only validation. Third, eight specific challenges and technical limitations were identified (Table 6), including the absence of studies in tropical maritime climates, the under-representation of predictive maintenance and cyber-security, and the lack of standardized DT platforms and communication protocol documentation. Fourth, the technical mapping of RQ4 reveals an overwhelmingly web-oriented communication layer (HTTP/REST, MQTT) and an almost complete absence of industrial protocols (OPC-UA, BACnet under-represented), confirming a research/IT positioning of the literature rather than a field-engineering one.

Future research should prioritize: (i) experimental validation of DT architectures in under-represented climates, particularly tropical-maritime, ultra-marine, and coastal conditions where humidity dynamics and salt-induced corrosion are dominant operational stressors; (ii) the development of standardized benchmarks comparable to Co-zyBench (Ma et al. [23]) for cross-study comparability; (iii) a dedicated research line on DT-aware predictive maintenance for HVAC systems exposed to challenging environments; and (iv) systematic engagement with cyber-security in IoT-centric DT architectures. These directions converge on a clear opportunity: piloting a Digital Twin implementation in a representative tropical or ultra-marine building would directly address several of the challenges identified above and contribute primary evidence to a literature that currently has very little of it.

6. References

References are formatted following IEEE style (sequential numbering by first appearance in the text). Numbers in square brackets after each entry correspond to internal corpus identifiers preserved for traceability with the screening artefacts (`included_papers_v4_classified.csv`). All DOIs are reproduced verbatim from the source databases.

6.1 Methodological references

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6.2 Included studies — IEEE format (n = 45)

Internal corpus identifiers (in the form [corpus-ID]) are kept after the IEEE entry for traceability with the project’s screening artefacts.

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